

Study Guide 3: Quantitative Tools

Fixed Income Study Guides

The same three tasks run through every tool: price a strip, compute its yield and duration, and do one bootstrap step. Calculators handle single bonds; spreadsheets and code build whole curves.

Common inputs used below:

- **Strip:** 20-year zero, settles Nov 15 2026, matures Nov 15 2046, yield 4.5%. Answer: price 41.065, modified duration 29.34 (30-year) or 19.56 (20-year).
- **Bootstrap step:** the 2.0-year bond from Study Guide 2. Coupon 4.6%, price 100, earlier coupons worth 6.6157. Answer: spot rate 4.6117%.

The calculations each tool performs:

$$P = \frac{100}{(1 + y/2)^n} \quad D^* = \frac{T}{1 + y/2} \quad s = 2 \left[\left(\frac{100 + c/2}{P - PV_{known}} \right)^{1/n} - 1 \right]$$

1. TI BA II Plus Professional

Strip price with TVM

1. 2nd CLR TVM to clear.
2. 2nd P/Y 2 ENTER (C/Y follows to 2), then 2nd QUIT.
3. 40 N · 4.5 I/Y · 0 PMT · 100 FV.
4. CPT PV → **-41.065**.

With P/Y = 2, I/Y is the annual rate and N is the number of half-years. 20 2nd xP/Y N also enters 40.

Strip price, yield and duration with the Bond worksheet

1. 2nd BOND, then 2nd CLR WORK.
2. SDT = 11.1526 ENTER (dates are MM.DDYY) ↓
3. CPN = 0 ENTER ↓ RDT = 11.1546 ENTER ↓ RV = 100 ENTER ↓
4. ACT (toggle with 2nd SET if it shows 360) ↓ 2/Y ↓
5. YLD = 4.5 ENTER ↓ PRI CPT → **41.065** ↓ AI → 0 ↓ DUR → **19.56** (modified duration, Professional model only).

To solve for yield instead, enter PRI and CPT at YLD.

Bootstrap step

1. Value the earlier coupons: $2.3 \times (0.980392 + 0.959267 + 0.936719) = 6.6157$. Keep discount factors in memory with STO 1-9 and RCL.
2. With P/Y = 2: 4 N · 93.3843 +| - PV · 0 PMT · 102.3 FV.
3. CPT I/Y → **4.6117**.

2. HP 12C (RPN)

Strip price with TVM

1. f CLEAR FIN.
2. 40 n · 4.5 ENTER 2 ÷ i (the 12C uses the rate per period) · 0 PMT · 100 FV.
3. PV → **-41.065**.

Bond functions (semiannual, Actual/Actual only)

1. g M.DY once to set US date format (MM.DDYYYY).
2. 4.5 i (annual yield) · 0 PMT (annual coupon %).
3. 11.152026 ENTER 11.152046 f PRICE → **41.065**; x≥y shows accrued interest.
4. For yield: price in PV, coupon in PMT, the two dates, then f YTM.

The 12C has no duration key. Compute modified duration for a zero by hand: 20 ENTER 1.0225 ÷ → 19.56.

Bootstrap step

1. 4 n · 93.3843 CHS PV · 0 PMT · 102.3 FV.
2. i → 2.3059 (per half-year), then 2 × → **4.6117**.
3. Or directly: 102.3 ENTER 93.3843 ÷ 4 1/x y× 1 – 2 × → 0.046117.

Forward rate from 1.0 to 1.5 years: 0.959267 ENTER 0.936719 ÷ 1 – 2 × → 0.0481.

3. Excel

Task	Formula	Result
Strip price	=PRICE("2026-11-15", "2046-11-15", 0, 4.5%, 100, 2, 1)	41.065
Strip price, direct	=100/(1+4.5%/2)^40	41.065
Yield from price	=YIELD("2026-11-15", "2046-11-15", 0, 41.065, 100, 2, 1)	4.50%
Modified duration	=MDURATION("2026-11-15", "2046-11-15", 0, 4.5%, 2, 1)	19.56

Arguments: frequency 2 = semiannual, basis 1 = Actual/Actual. Use cell references or DATE() in real models.

Bootstrap layout (one row per half-year; row 2 is the first bond). Columns: A = period n, C = coupon in percent units (4.2, not 4.2%), D = price, E = discount factor, F = spot, G = forward.

Cell	Formula (fill down from row 3)
E2	=D2/(100+C2/2)
E3	=(D3 - (C3/2)*SUM(E\$2:E2))/(100+C3/2)
F2	=2*(E2^(-1/A2) - 1)
G3	=2*(E2/E3 - 1)

For a smooth fitted curve, put Nelson-Siegel parameters in cells and let Solver minimize squared pricing errors.

4. Python

Library	Role	License
QuantLib (pip install QuantLib)	Industry standard. Wraps the C++ library; full day counts, calendars, bond helpers, curve bootstrapping and interpolation	Open source, BSD-style; free for commercial use

Library	Role	License
rateslib	Modern, Pythonic curve building and bond/swap analytics with automatic differentiation	Source-available: free for personal and educational use; commercial use needs a paid licence
FinancePy	Readable pure-Python library, good for teaching	Open source
numpy-financial, SciPy, pandas	Basic TVM, root-finding and curve fitting for hand-built models	Open source

QuantLib bootstrap of Example B (tested; matches the hand results):

```
import QuantLib as ql
today = ql.Date(15, ql.January, 2026)
ql.Settings.instance().evaluationDate = today
cal, dc = ql.NullCalendar(), ql.Thirty360(ql.Thirty360.BondBasis)
helpers = []
for months, cpn in [(6, 4.0), (12, 4.2), (18, 4.4), (24, 4.6)]:
    sched = ql.Schedule(today, today + ql.Period(months, ql.Months), ql.Period(ql.Semiannual),
                        cal, ql.Unadjusted, ql.Unadjusted, ql.DateGeneration.Backward, False)
    helpers.append(ql.FixedRateBondHelper(ql.QuoteHandle(ql.SimpleQuote(100.0)),
                                         0, 100.0, sched, [cpn / 100], dc))
curve = ql.PiecewiseLogLinearDiscount(today, helpers, dc)
print(curve.zeroRate(2.0, ql.Compounded, ql.Semiannual).rate()) # 0.046117
```

30/360 makes every period exactly half a year so the output matches the hand example. Real Treasury work uses `ql.ActualActual(ql.ActualActual.Bond)` and a US government calendar.

5. Julia

- **FinanceModels.jl** (JuliaActuary, formerly Yields.jl) is the main maintained package. It takes market quotes such as ZCBYield, ParYield or CMTYield and fits curves with `fit(model, quotes, Fit.Bootstrap())`.
- Models include bootstrapped splines, Nelson-Siegel, Nelson-Siegel-Svensson and Smith-Wilson. Functions `discount`, `zero`, `forward` and `par` read rates off the fitted curve.
- **ActuaryUtilities.jl** adds duration, convexity and present-value helpers.

Plain-Julia bootstrap (same logic as the tested Python; not run here):

```
coupons = [4.0, 4.2, 4.4, 4.6]; prices = fill(100.0, 4)
spots = Float64[]
for n in eachindex(coupons)
    cpn = coupons[n] / 2
    pv_known = sum((cpn / (1 + spots[k] / 2)^k for k in 1:n-1); init = 0.0)
    push!(spots, 2 * (((100 + cpn) / (prices[n] - pv_known))^(1 / n) - 1))
end
println(spots) # 0.04, 0.042021, 0.044059, 0.046117
```

6. Choosing a tool

Tool	Best for	Bootstrap a whole curve?	Duration built in?
TI BA II Plus Professional	Exams, single-bond checks	One step at a time, by hand	Yes (modified)
HP 12C	Exams, quick RPN checks	One step at a time, by hand	No
Excel	Transparent teaching models, small curves	Yes, with fill-down formulas	Yes (DURATION, MDURATION)
Python + QuantLib	Production-grade curves and pricing	Yes, with interpolation choices	Yes (BondFunctions)
Julia + FinanceModels.jl	Fast actuarial and curve modeling	Yes, including fitted models	Yes (ActuaryUtilities.jl)

The CFA exams allow only the TI BA II Plus (including Professional) and HP 12C families, which is why both appear here.

7. Practice set

1. On either calculator, price a 10-year strip at 4.0%. *67.297*.
2. On the TI Bond worksheet, find the yield of a strip settling 11/15/2026, maturing 11/15/2041, priced at 50. *4.67%*.
3. In Excel, extend the bootstrap table with a 2.5-year 4.8% par bond. *Spot $\approx$ 4.82%*.
4. In Python, change the QuantLib day count to Actual/Actual and explain why the rates move slightly. *Periods are no longer exactly 0.5 years*.

8. CFA-style review questions

Original questions in the CFA Level I format (three choices, one correct). Not CFA Institute material. Solve each on the TI BA II Plus or HP 12C.

1. A 10-year Treasury note pays a 4% coupon semiannually and yields 5%. Its price per 100 is closest to:
 - A. 92.21
 - B. 92.28
 - C. 107.79
2. A 5-year bond pays a 6% coupon semiannually and yields 5%. Its price per 100 is closest to:
 - A. 104.33
 - B. 104.38
 - C. 105.00
3. The HP 12C's bond functions (f PRICE, f YTM) assume:
 - A. semiannual coupons and an Actual/Actual day count
 - B. annual coupons and a 30/360 day count
 - C. quarterly coupons and an Actual/360 day count
4. To compute a bond's yield to maturity from settlement and maturity dates in Excel, the most appropriate function is:
 - A. YIELD
 - B. RATE
 - C. IRR
5. On the TI BA II Plus, a student solves a semiannual bond with P/Y left at 12. The computed I/Y will most likely be:
 - A. correct, because P/Y affects only the display
 - B. wrong, because the calculator assumes monthly periods
 - C. correct only if the bond trades at par

Answer key

1. A. $N = 20$, $I/Y = 2.5$ per period, $PMT = 2$, $FV = 100 \rightarrow PV = 92.21$. B uses annual compounding; C is a premium price, impossible when the yield exceeds the coupon.
2. B. $N = 10$, $I/Y = 2.5$, $PMT = 3$, $FV = 100 \rightarrow 104.38$. A uses annual compounding.
3. A. The 12C bond functions are fixed to U.S. Treasury conventions.
4. A. YIELD takes dates, coupon, price, redemption, frequency and basis. RATE needs whole periods; IRR needs a cash flow list.
5. B. P/Y and C/Y must match the bond's payment frequency (2).

9. Bloomberg Terminal (draft: verify on the campus Terminal)

The Terminal is a sixth tool alongside the ones above, and it connects to Excel and Python.

Command	What to do	Check against this guide	Verified
YAS <GO>	The Terminal's bond calculator: enter a price or yield, read price, yield, duration and convexity	Use it to check TI and HP 12C answers	☐
Excel add-in	Bloomberg tab → Spreadsheet Builder; functions BDP (current value), BDH (history), BDS (lists)	Rebuild the section 3 Excel table with live data	☐
FLDS <GO>	Field search: find the exact field names for price, yield, duration, accrued interest	Field names used in these guides	☐
BQL	Bloomberg Query Language in Excel, e.g. <code>=BQL("CT10 Govt","yield")</code> (confirm syntax)	Bulk pulls for curves and ladders	☐
BQNT <GO>	BQuant: Python notebooks on the Terminal	Run the section 4 Python examples against live data	☐
BMC <GO>	Bloomberg Market Concepts course, including a fixed income module	Student onboarding before these exercises	☐

The `blpapi` Python package (Desktop API) works only on a machine logged in to the Terminal; licensing limits redistributing the data.

Exercise: pull the current 10-year note's price, yield and modified duration with BDP, then reproduce each value with PRICE, YIELD and MDURATION in the same workbook.

10. Reading in Bodie, Kane & Marcus, *Investments*, 13th ed. (2024)

Section	Topic in the book	Supports
14.2 Bond Pricing	Bond Pricing between Coupon Dates; Excel PRICE spreadsheet	Excel bond pricing
14.3 Bond Yields	Yield to Maturity; Yield to Call	Excel YIELD and calculator yield solving
16.1 Interest Rate Risk	Duration; Spreadsheet 16.1	Duration built cash flow by cash flow in Excel
5.1 Measuring Returns over Different Holding Periods	Annual Percentage Rates; Continuous Compounding	Compounding conversions (ICONV, EFFECT)

The book's tool coverage is Excel only, through its Excel Applications and the spreadsheet templates on its student site (mhhe.com/Bodie13e). The calculator keystrokes, Python and Julia material in this guide go beyond the text.

Sources: [rateslib licence](#) · [FinanceModels.jl documentation](#) · [FinanceModels.jl on GitHub](#)